

SMART AGRICULTURAL SOIL-MOISTURE MONITORING GATEWAY

Using Linux Device Driver, System Programming and C++

1. Project Introduction

1.1 Project Overview

The Smart Agricultural Soil-Moisture Monitoring Gateway is a Linux-based software prototype designed to monitor soil-moisture conditions from multiple sensor nodes.

The system demonstrates the integration of Linux device-driver development, hardware abstraction, system programming, C++, backend services, database storage and web-based monitoring.

The current prototype uses simulated sensor nodes so that the complete software architecture can be developed and tested without requiring physical agricultural sensor hardware.

1.2 Problem Statement

Agricultural irrigation requires information about soil conditions to avoid unnecessary watering and to identify dry areas.

A multi-node monitoring system can collect sensor information from different locations and process it through a central gateway.

The project addresses this requirement by designing a Linux-based gateway capable of processing sensor information, classifying soil conditions and providing monitoring information through a web dashboard.

1.3 Objectives

The main objectives are:

- Develop a Linux-based agricultural monitoring gateway.
- Demonstrate Linux device-driver interaction.
- Implement a hardware abstraction layer.
- Process sensor information using C++.
- Support multiple sensor nodes.
- Detect dry-soil conditions.
- Generate irrigation recommendations.
- Store historical sensor information.
- Provide REST APIs.

- Develop a web-based monitoring dashboard.
 - Demonstrate complete system integration.
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2. Scope

The project includes:

- Simulated multi-node soil-moisture sensors.
- Linux device-driver interface.
- Hardware Abstraction Layer.
- C++ gateway and sensor processing.
- Python Flask backend.
- SQLite database.
- REST API endpoints.
- Monitoring dashboard.
- Sensor analytics.
- Git-based version control.

The current prototype does not include physical sensors or automatic irrigation hardware.

3. System Architecture

The final architecture is:

“text Simulated Sensor Nodes | v Hardware Abstraction Layer | v Linux Device Driver | v Gateway Daemon | v C++ Sensor Processor | v Flask Backend | v SQLite Database | v Monitoring Dashboard